MagicBox Installation Guide

Goal: Portainer reachable from the LAN at **https://magicbox.local:9443/**. MagicBox has no fixed IP: it is always reached as `magicbox.local` (mDNS via Avahi). `magicbox$` = on the Mini PC, `controller$` = on your workstation in this repository. Replace `<...>` with your values.

1. Manual Steps (Mini PC)

1.1 Before you start

- [ ] Back up anything on the internal disk (the OS install erases it).
- [ ] Have: USB stick (4 GB+), Ethernet cable to the LAN, router admin access.
- [ ] Controller and Mini PC on the same subnet (not guest Wi-Fi).

1.2 Firmware (UEFI)

Setup key is usually `Del`, boot menu `F7` (check the boot screen).

- [ ] Apply a GMKtec firmware update for your model, if one exists.
- [ ] Boot mode: UEFI. Secure Boot: enabled.
- [ ] *Restore on AC Power Loss*: **Power On**.
- [ ] Save and exit.

1.3 Install Ubuntu Server 24.04 LTS

(Also supported: Ubuntu 26.04, Debian 12/13.)

- [ ] Download the ISO from `https://ubuntu.com/download/server`, verify its SHA-256, write it to the USB stick, boot from it.
- [ ] Network: wired, DHCP (no static IP).
- [ ] Storage: entire internal disk, **no** passphrase encryption (it blocks unattended reboots).
- [ ] Server name: **`magicbox`**. Create your admin user (`<admin-user>`).
- [ ] Tick **Install OpenSSH server**. Featured snaps: **none**.
- [ ] Reboot, remove the USB stick, log in.

1.4 Update and collect facts

```
magicbox$ sudo apt update && sudo apt full-upgrade -y && sudo reboot
magicbox$ hostname                          # must print magicbox
magicbox$ cat /etc/os-release; uname -m     # supported OS, x86_64
magicbox$ ip -br link                       # LAN interface name
magicbox$ ip -br -4 addr                    # LAN subnet (e.g. 192.168.1.x/24)
```

1.5 Avahi (magicbox.local)

```
magicbox$ sudo apt install -y avahi-daemon libnss-mdns
magicbox$ sudo systemctl enable --now avahi-daemon
magicbox$ avahi-resolve-host-name -4 magicbox.local
```

- [ ] The last command prints `magicbox.local` and an IPv4 address.
- [ ] From the controller: `ping -c 3 magicbox.local` answers. (Linux controller: `sudo apt install avahi-daemon libnss-mdns` first.)

1.6 SSH key

```
controller$ ssh-keygen -t ed25519            # skip if you have a key
controller$ ssh-copy-id <admin-user>@magicbox.local
controller$ ssh <admin-user>@magicbox.local 'sudo -v && echo sudo-ok'
```

- [ ] Last command prints `sudo-ok`.
- [ ] Optional, after key login works (keep a session open while testing):

```
magicbox$ echo 'PasswordAuthentication no' | sudo tee /etc/ssh/sshd_config.d/10-magicbox.conf
magicbox$ sudo sshd -t && sudo systemctl reload ssh
```

1.7 Values for the inventory

Value	Source	Yours
ansible_host	always magicbox.local	magicbox.local
ansible_user	admin user	_____
magicbox_lan_interface	ip -br link	_____
magicbox_lan_cidr	LAN subnet, e.g. 192.168.1.0/24	_____

2. Controller setup

- [ ] Install `uv` (`brew install uv`), `git`, `make`. Windows: use WSL2.

```
controller$ make setup
controller$ make check
```

3. Configuration

```
controller$ cp inventory/hosts.example.yml inventory/hosts.yml
controller$ uv run ansible-vault create inventory/group_vars/magicbox/vault.yml
```

- [ ] In `inventory/hosts.yml`: set `ansible_user`, `magicbox_lan_interface`, `magicbox_lan_cidr` (1.7). Keep `ansible_host: magicbox.local`.
- [ ] In `vault.yml`: `vault_portainer_admin_password: "<12+ chars>"`. Keep the vault password in your password manager.

```
controller$ make secrets-check
controller$ uv run ansible magicbox -m ansible.builtin.ping    # expect "pong"
```

4. Deployment

```
controller$ make deploy EXTRA="-K -e magicbox_confirm_destructive=true"   # first run
controller$ make deploy EXTRA="-K"                                        # later runs
```

`-K` asks for the sudo password. The first run needs the confirmation flag because it enables the firewall. The CA certificate is saved to `artifacts/magicbox-local-ca.crt`.

5. Verification

```
controller$ make verify EXTRA="-K"
controller$ make idempotency VAULT_PASSWORD_FILE=<file> EXTRA="--become-password-file=<file>"
```

- [ ] `make verify` passes.
- [ ] `make idempotency` ends with "Idempotency check passed". Keep the password files outside the repo with mode 0600; delete them afterwards.
- [ ] From another LAN computer: `ping magicbox.local` works.
- [ ] Browser (CA trusted, section 6): `https://magicbox.local:9443/` opens with no warning; log in as `admin` with the vault password.
- [ ] `sudo reboot` on MagicBox: the page comes back, settings kept.
- [ ] `magicbox$ sudo ss -ltn | grep -E ':(9000|2375|2376) '` prints nothing.

6. TLS trust (each client, once)

- [ ] Check the fingerprints match:

```
controller$ openssl x509 -in artifacts/magicbox-local-ca.crt -noout -fingerprint -sha256
magicbox$ sudo openssl x509 -in /opt/magicbox/portainer/certs/ca.crt -noout -fingerprint
-sha256
```

- [ ] Copy `magicbox-local-ca.crt` to the client and install it:

Client	Command / steps
macOS	sudo security add-trusted-cert -d -r trustRoot -k /Library/Keychains/System.keychain magicbox-local-ca.crt
Windows (admin PowerShell)	Import-Certificate -FilePath magicbox-local-ca.crt -CertStoreLocation Cert:\LocalMachine\Root
Ubuntu/Debian	sudo cp magicbox-local-ca.crt /usr/local/share/ca-certificates/ && sudo update-ca-certificates
Firefox	Settings > Certificates > View Certificates > Authorities > Import > trust for websites
iOS	Install the profile, then General > About > Certificate Trust Settings > enable
Android	Settings > Security > Encryption & credentials > Install a certificate > CA certificate

- [ ] Restart the browser.

The CA only covers `magicbox.local` and `magicbox`. The server certificate is renewed automatically by any playbook run within 30 days of expiry.

6.1 If a client cannot resolve magicbox.local

- [ ] Client on the same LAN subnet (not guest Wi-Fi, not VPN, not another VLAN).
- [ ] macOS, iOS, Windows 10/11: built in. Android 12+: most browsers.
- [ ] Linux: `sudo apt install avahi-daemon libnss-mdns`.
- [ ] On MagicBox: `systemctl status avahi-daemon` is active.

7. Troubleshooting

Symptom	Fix
"Unsupported OS"	Install Ubuntu Server 24.04 LTS
magicbox_lan_interface assertion	Use a name listed in the error
Admin password assertion	Create vault.yml; run with --ask-vault-pass
"ufw is inactive"	Add -e magicbox_confirm_destructive=true
Controller "outside" the LAN	Fix magicbox_lan_cidr or run from the LAN
Conflicting Docker packages	Check nothing uses them, then add the confirmation flag
Docker key fingerprint mismatch	Stop and verify Docker's key out of band
magicbox.local not resolving	systemctl status avahi-daemon; see 6.1
Name became magicbox-2.local	Rename the other magicbox device; restart avahi-daemon
Certificate warning	Install the CA (section 6); restart the browser
Connection refused	sudo docker ps -a; sudo docker logs portainer; sudo ufw status
Vault password rejected at login	Portainer was already initialised: reset (8.3)

8. Recovery

8.1 Repair drift

```
controller$ make deploy EXTRA="-K"
```

8.2 Firewall lockout

At the console: `sudo ufw disable`, fix `magicbox_lan_cidr` (IPv4 subnet), then redeploy with `-e magicbox_confirm_destructive=true`.

8.3 Lost admin password

```
magicbox$ sudo docker stop portainer
magicbox$ sudo docker run --rm -v portainer_data:/data portainer/helper-reset-password
magicbox$ sudo docker start portainer
```

8.4 Backup and restore

```
magicbox$ sudo docker stop portainer
magicbox$ sudo tar -C / -czf /root/magicbox-backup-$(date +%F).tgz opt/magicbox
var/lib/docker/volumes/portainer_data
magicbox$ sudo docker start portainer
```

The archive contains the CA key: store it encrypted, off the box. To restore, reinstall (sections 1-4), stop Portainer, run `sudo tar -C / -xzf <archive>`, and start Portainer.

8.5 Rotate the CA

Delete `/opt/magicbox/portainer/certs/` on MagicBox, run `make deploy`, and install the new CA on every client. Remove the old CA from the clients.